

Plan Urena

The MK3 pan-tilt: 3D-printed, belt-driven, on hollow bearings. What to buy, what to print, how to wire it and how to build it.



01 The plan

02 The design

03 What to buy

04 Reuse from MK2

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06 Camera options

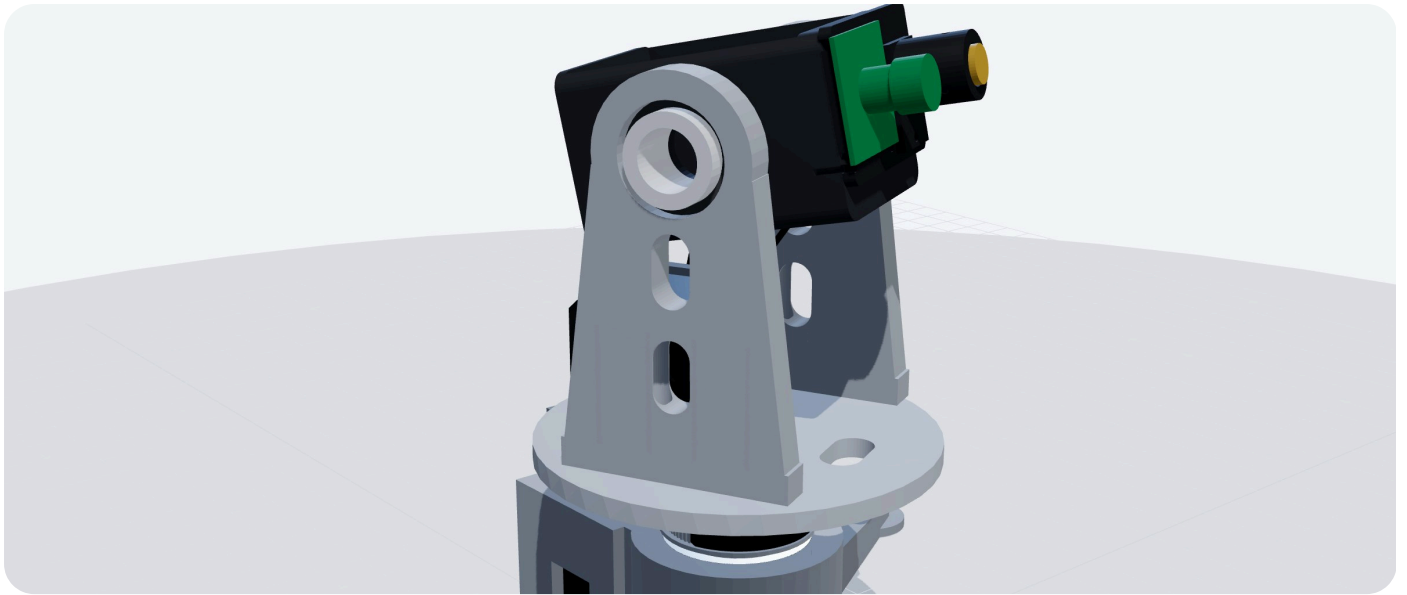
07 Wiring and power

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The plan



A stiff, belt-driven pan-tilt for the tracker, printed in PETG and built around big hollow bearings. It uses the same software and the same protocol as MK2. The motors, drivers, sensors and camera come from MK2.

DRIVE

4:1 GT2 belts

on both axes

FINE STEPS

35.6

motor steps per degree

TRAVEL

pan $\pm 150^\circ$ · tilt $\pm 40^\circ$

clear to -45° / $+60^\circ$ in CAD

BEARINGS

2 × 6808 + 2 × 6806

hollow: cables go through

PRINTED

20 files · $\approx 1,074$ g

PETG

CONTROLLER

UNO + CNC Shield V3

no breadboard, fused 12 V

TO BUY

₹4,356–7,086

incl. spares

PRINTING

₹5,370–10,740

at ₹5–10 per gram

What changes from MK2

	MK2	PLAN URENA
Frame	Acrylic, L-brackets, couplings	3D-printed PETG, designed to fit
Drive	Motor shafts carry the load	4:1 belts on both axes: 4 × torque, 4 × finer
Bearings	None of their own	Big hollow bearings: stiff, no wobble
Tilt motor	On the moving head frame	On the deck, under the head: compact and balanced
Cables	Loose loop around the rig	Down the hollow middle
Wiring	Breadboard + loose jumpers	UNO + CNC shield: everything plugs in
12 V power	Adapter pigtail into a breadboard	Panel jack → 3 A fuse → switch → screw terminal
Pan sensor	On the output	On the pan motor shaft: 4 × finer reading

Three ways to build a pan-tilt, and why we chose belts

	PRINTED GEARS	DIRECT DRIVE (MOTOR = AXIS)	BELTS 4:1 + BEARINGS (PLAN URENA)
Seen in	Hobby webcam trackers with big printed spur gears	Our MK1 / MK2	Compact belt pan-tilts with a tripod base and the tilt motor on the turntable
Accuracy	Printed teeth have 0.3–1° of play (backlash): the aim wobbles when it reverses	Limited by the motor step and any load on its shaft	No play: a tensioned GT2 belt has no backlash, and 4:1 makes each step 4× finer
Stability	Good if the frame is stiff	The motor shaft carries the whole load	Best: big hollow bearings carry the load, the motors only turn
Parts to buy	Fewest (no belts, no pulleys)	Fewest	+ 4 bearings, 2 belts, 2 metal pulleys (≈₹1,000)
Easy to build	Easy, but gear teeth must print cleanly	Easiest	Easy: slide the motor to tighten each belt, no fine gear meshing

We keep belts: the same layout as the compact belt pan-tilt we liked (tripod base, pan motor in the base, tilt motor standing on the turntable, belt up the side), which gives the most accuracy for the least effort.

The plan in six steps

- 1. Buy**

One trip to Amar Robotics and a bearing shop, plus one online order for belts.
- 2. Print the coupon**

One small print. Check the fits (page "Tests").
- 3. Print everything**

≈1,074 g, about 60–80 printer-hours at normal speed.
- 4. Assemble**

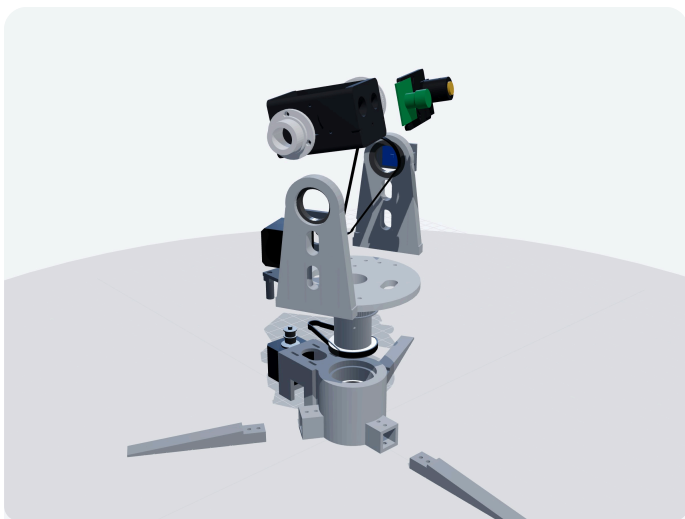
14 steps, about one day.
- 5. Wire + flash**

Half a day. Nothing to solder.
- 6. Test + tune**

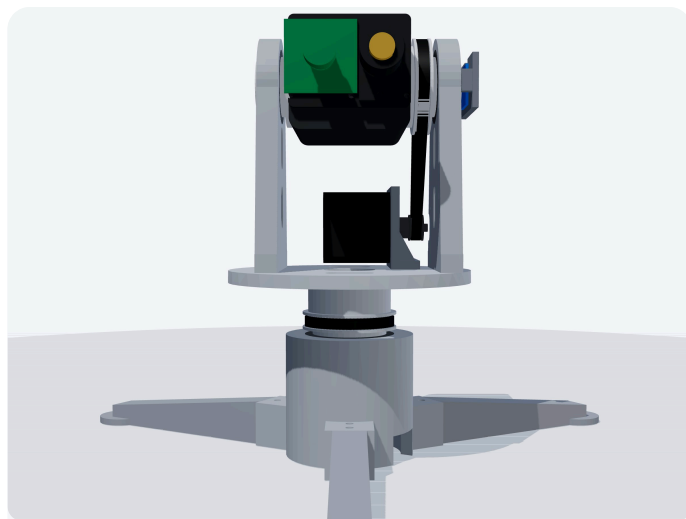
Self-test, belt tension, laser aim, then the Live page.

Keep it internal. "Plan Urena" and "MK3" stay on these build pages and PDFs only. Not on the public website, the deck, the video or any official material.

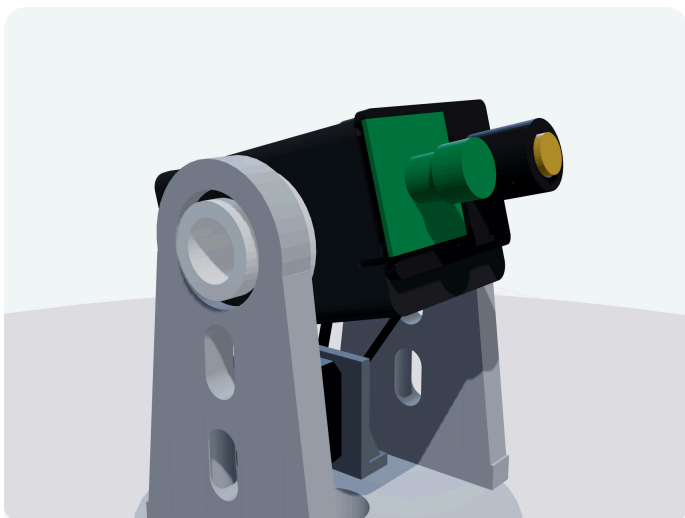
The design



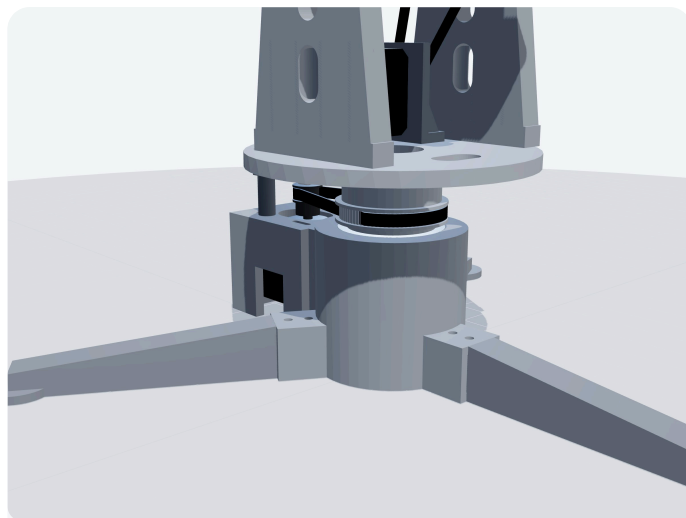
Exploded: how the parts stack



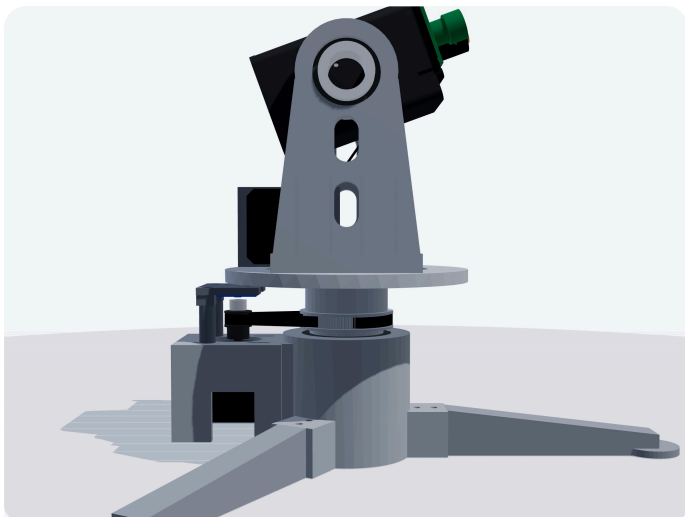
Front



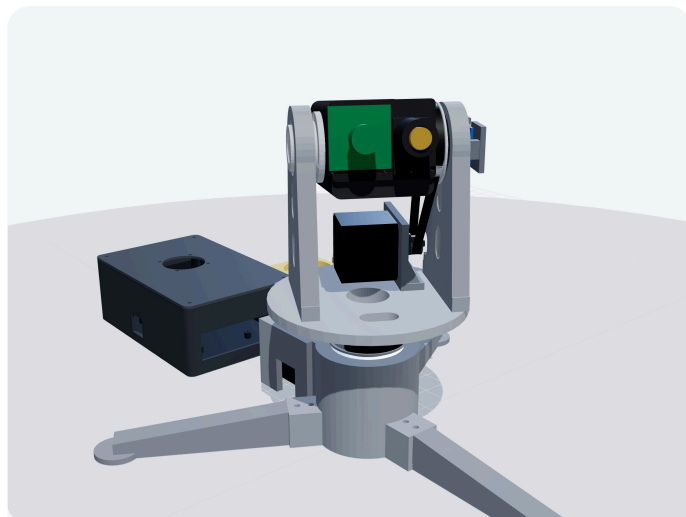
Head and tilt drive



Base, pan belt and pan motor



Side



With the electronics box

What to buy

Prices are estimates. Show this list at the shop and compare. **Spares are included** (a second UNO, CNC shield, drivers, belts, a pulley and a laser) so a dead part never costs a day.

PARTS TO BUY

₹4,356–7,086

with spares

PRINTING

₹5,370–10,740

≈1,074 g PETG

TOTAL

₹9,726–17,826

MK2 parts reused

OPTIONAL CAMERA

₹2,100–4,100

see "Camera"

Where. **Amar Robotics** for electronics, pulleys and the laser. A **bearing shop** for the four bearings: say "6808 2RS, two" and "6806 2RS, two". A **hardware shop** for screws. **Online** (Robu, Amazon) for the closed belts, which local shops rarely stock. If a shop has no 232 mm belt, a 240 mm belt fits too (anything 227–245 mm).

Bearings, belts, pulleys

₹1,100–1,970

The moving parts. Take the test coupon and the printed pins to the bearing shop.

ITEM	QTY	EACH	TOTAL	WHERE
6808-2RS bearing (40 × 52 × 7 mm) 6808-ZZ is the same size (metal shields). Any brand.	2	₹150–250	₹300–500	Bearing shop
6806-2RS bearing (30 × 42 × 7 mm) 6806-ZZ is the same size.	2	₹120–200	₹240–400	Bearing shop
GT2 pulley, 20 teeth, 5 mm bore, for 6 mm belt 1 spare. Metal (aluminium), with 2 grub screws.	3	₹80–150	₹240–450	Amar Robotics
GT2 closed belt 6 mm wide, 232 mm 1 spare. 240 mm also fits (anything 227 to 245 mm).	2	₹80–150	₹160–300	Online (Robu / Amazon)
GT2 closed belt 6 mm wide, 280 mm 1 spare. 276 to 284 mm fits.	2	₹80–160	₹160–320	Online (Robu / Amazon)

Controller and power

₹2,570–3,850

No breadboard. Everything plugs into the CNC shield.

ITEM	QTY	EACH	TOTAL	WHERE
Arduino UNO R3 (CH340 clone is fine) + USB-B cable 1 spare. The MK2 Nano is a third backup.	2	₹450–700	₹900–1,400	Amar Robotics
CNC Shield V3 (for UNO) 1 spare. Must say "V3" and fit on an UNO.	2	₹250–350	₹500–700	Amar Robotics
A4988 driver with heatsink Spares. Reuse the 2 from MK2 if they still work.	2	₹150–200	₹300–400	Amar Robotics
12 V 5 A adapter (SMPS), 5.5 × 2.1 mm plug 12 V 3 A is the minimum. Not a 12 V 1 A phone-style one.	1	₹450–650	₹450–650	Amar Robotics

Panel DC jack 5.5 × 2.1 mm (female, with nut) Screws into the box wall.	1	₹30–50	₹30–50	Amar Robotics
Fuse holder (5 × 20 mm) + 3 A fuses Buy a pack of 5 fuses.	1	₹40–80	₹40–80	Amar Robotics
Rocker switch (KCD1, 6 A) Any 2-pin ON/OFF switch rated 3 A or more.	1	₹20–40	₹20–40	Amar Robotics
Silicone wire 18 AWG, red + black 1 m of each colour. For the 12 V only.	2	₹50–80	₹100–160	Amar Robotics
40 mm fan, 12 V Cools the two drivers.	1	₹80–120	₹80–120	Amar Robotics
MOSFET switch module (D4184 / AOD4184) Switches the vibration motor. A 2N2222 + diode from the MK2 kit also works.	1	₹60–90	₹60–90	Amar Robotics
Dupont jumper cables, female-female, 20 cm One pack of 40. For the sensors.	1	₹60–100	₹60–100	Amar Robotics
Spiral cable wrap, 1 m Keeps the cable bundle neat.	1	₹30–60	₹30–60	Hardware shop

Laser and magnets

₹260–520

ITEM	QTY	EACH	TOTAL	WHERE
12 mm laser module, 650 nm red, 5 mW, 5 V 1 spare. Only 5 mW (Class 2/3R). Never a "burning" laser.	2	₹100–200	₹200–400	Amar Robotics
Diametric magnet 6 × 2.5 mm (AS5600 type) Spares. One usually comes with each AS5600 board.	2	₹30–60	₹60–120	Online (Robu / Amazon)

Screws and small things

₹426–746

Buy an M3 kit; the counts are in the table on the "Assemble" page.

ITEM	QTY	EACH	TOTAL	WHERE
M3 screw kit (6 to 30 mm) with nuts and washers	1	₹250–400	₹250–400	Hardware shop
M3 × 35 mm screws For the pan sensor bridge (2 + 2 spare).	4	₹5–10	₹20–40	Hardware shop
M4 × 30 mm bolts + nuts Legs.	6	₹8–15	₹48–90	Hardware shop
M4 × 20 mm bolts + nuts + washers Arms to the deck.	8	₹6–12	₹48–96	Hardware shop
M2.5 × 6 mm screws Sensor boards. M2 × 6 self-tapping also works.	6	₹5–10	₹30–60	Hardware shop
Super glue (CA) One drop fixes the magnet cap.	1	₹30–60	₹30–60	Hardware shop

Do not buy: a breadboard, a 12 V 1 A adapter, a "CNC Shield V4" (the Nano one has known design faults), or any laser stronger than 5 mW.

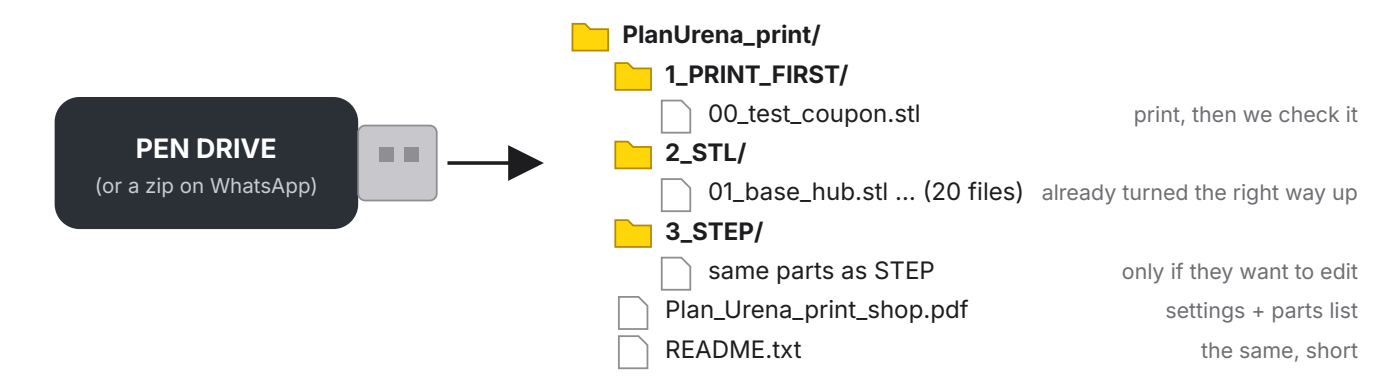
Reuse from MK2

Most of the electronics come straight from MK2. The frame is new.

MK2 PART	REUSE?	NOTES
2 x NEMA17 motors + their cables	Yes	Same motors, same 4-pin cables.
2 x A4988 drivers	Test first	Plug each into the CNC shield and run the self-test. If one burned with the breadboard, use a spare.
2 x AS5600 sensors + magnets	Yes	Pan sensor now reads the pan motor shaft; tilt sensor reads the end of the right trunnion.
TCA9548A sensor splitter	Yes	Still needed: both AS5600s have the same address.
USB camera	Yes	Works as it is. See "Camera" for a sharper option.
KY-008 laser	Backup	The head is made for a 12 mm tube laser. Keep the KY-008 as a spare.
Vibration motor + transistor kit	Yes	Or use the MOSFET module.
Arduino Nano	Spare board	The MK3 firmware also runs on the Nano (set BOARD_CNC_SHIELD to 0).
Beacon + decoy	Yes	No change.
Laptop software + Live page	Yes	Pick "Both belts 20T → 80T (4:1)" in the drive menu.
12 V 2 A adapter	Tests only	Too weak for two motors at full speed, and its wire was damaged. Keep it for bench tests of one motor.
Breadboard, acrylic, L-brackets, couplings, servo	No	Not needed any more.

Printed parts

Give the print shop the files exactly as they are: every STL is already turned the right way up on the bed. Print the **test coupon first**, check it, then print the rest.

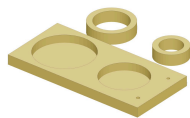


What to hand over. The STL files are what they print. The STEP files are only for editing.

Printer settings (the same for every part)

Material	PETG (not PLA: PLA slowly bends under the belt pull and near the motors). Head, front plate and laser holder in black; the rest any colour.
Layer height	0.2 mm, 0.4 mm nozzle
Walls	4 walls, 5 top and 5 bottom layers
Infill	40 % gyroid
Supports	None, except inside the motor pocket of the base hub
Brim	On the base hub and the electronics box
Scale	100 %. Do not rescale anything.
Total	≈1,074 g PETG, about 60–80 printer-hours at normal speed (a fast printer takes about half). Cost at ₹5–10 per gram: ₹5,370–10,740. Ask for one price for the whole job.

The 20 pieces



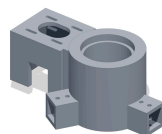
Test coupon

00_test_coupon.stl

Checks the fits before anything big is printed: both bearing pockets, both bearing pins, an M3 nut and a screw hole.

Qty 1
Size 130 × 110 × 10 mm
PETG ≈62 g
Colour Any
Supports None

PRINT THIS FIRST. We check it before the rest.

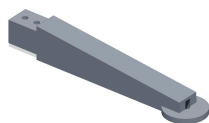


Base hub 01_base_hub.stl

Holds the two big pan bearings and the pan motor. The cables come out of a hole at the bottom.

Qty 1
Size 155 × 114 × 70 mm
PETG ≈195 g
Colour Grey
Supports Only inside the motor pocket

Biggest part. Use a brim.

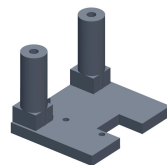


Leg 02_leg_1 / _2 / _3.stl

Three legs make a stable tripod. Bolt into the base hub.

Qty 3
Size 160 × 32 × 26 mm
PETG ≈126 g
Colour Grey
Supports None

Print 3 (files 02_leg_1, _2, _3 are the same leg).

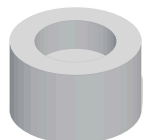


Pan sensor bridge

03_pan_sensor_bridge.stl

Holds the pan sensor board just above the pan motor shaft.

Qty 1
Size 35 × 42 × 31 mm
PETG ≈11 g
Colour Grey
Supports None



Magnet cap

04_magnet_cap.stl

Sits on the pan motor shaft tip and holds the small round magnet dead centre.

Qty 1
Size 10 × 10 × 6 mm
PETG ≈<1 g
Colour Any
Supports None

Tiny: print 2 (one spare).



Spindle washer

05_spindle_washer.stl

Three screws pull it up against the lower bearing so the pan has no play.

Qty 1
Size 43 × 43 × 3 mm
PETG ≈3 g
Colour Any
Supports None



Pan turntable

06_pan_turntable.stl

The hollow tube, the 80-tooth pulley and the round deck, all in one print.

Qty 1
Size 140 × 140 × 79 mm
PETG ≈142 g
Colour Grey
Supports None

Print deck-down (the file is already turned).



Right arm 07_arm_right.stl

Holds the right tilt bearing. The tilt pulley turns just inside it; the tilt sensor bracket screws to its outside.

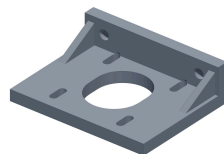
Qty 1
Size 80 × 132 × 12 mm
PETG ≈64 g
Colour Grey
Supports None



Left arm 08_arm_left.stl

Holds the left tilt bearing. The cables leave the head on this side.

Qty 1
Size 80 × 132 × 12 mm
PETG ≈64 g
Colour Grey
Supports None

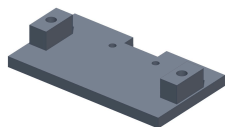


Tilt motor stand

09_tilt_motor_plate.stl

Stands on the deck under the head and holds the tilt motor. Slots let the motor slide to tighten the belt.

Qty 1
Size 62 × 48 × 17 mm
PETG ≈19 g
Colour Grey
Supports None

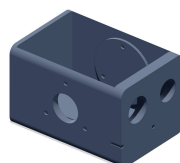


Tilt sensor bracket

10_tilt_sensor_bracket.stl

Screws to the outside of the right arm and holds the tilt sensor facing the magnet in the trunnion end.

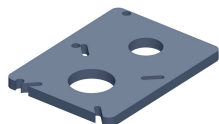
Qty 1
Size 54 × 30 × 10 mm
PETG ≈8 g
Colour Grey
Supports None



Head 11_head.stl

The box that carries the camera and the laser. Open top for easy wiring.

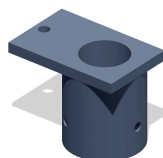
Qty 1
Size 92 × 64 × 58 mm
PETG ≈97 g
Colour Black
Supports None
Black stops stray light.



Front plate 12_front_plate.stl

The camera board screws to this. Slots fit 28 to 34 mm hole patterns.

Qty 1
Size 48 × 64 × 4 mm
PETG ≈13 g
Colour Black
Supports None

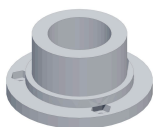


Laser holder

13_laser_holder.stl

Tube for a 12 mm laser, with 3 screws to aim it exactly at the camera centre.

Qty 1
Size 34 × 22 × 26 mm
PETG ≈9 g
Colour Black
Supports None

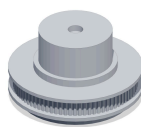


Left trunnion

14_trunnion_left.stl

Hollow tube bolted to the head. Turns in the left bearing. Cables pass through it.

Qty 1
Size 46 × 46 × 21 mm
PETG ≈14 g
Colour Grey
Supports None

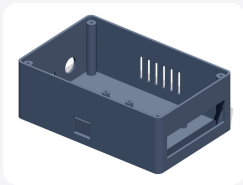


Right trunnion + tilt pulley

15_trunnion_right.stl

One print: bolts to the head, the 80-tooth tilt pulley, then the tube that turns in the right bearing. The tilt magnet sits in its end.

Qty 1
Size 54 × 54 × 31 mm
PETG ≈31 g
Colour Grey
Supports None



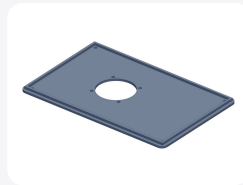
Electronics box

16_electronics_box.stl

Holds the Arduino UNO + CNC shield. Power jack, fuse and switch on the wall.

Qty 1
Size 170 × 110 × 60 mm
PETG ≈159 g
Colour Dark grey
Supports None

Use a brim.



Box lid 17_electronics_lid.stl

Lid with a 40 mm fan over the motor drivers.

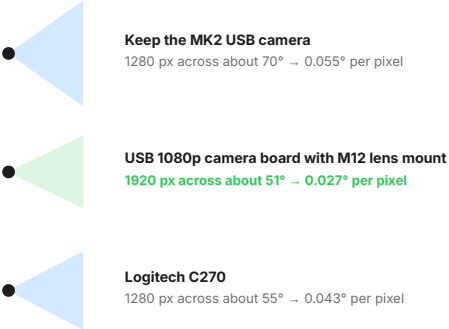
Qty 1
Size 170 × 110 × 5 mm
PETG ≈57 g
Colour Dark grey
Supports None

Camera options

Accuracy comes from the mechanics and from degrees per pixel, not from a fancy camera. The belts and bearings already give the big gain. A narrower lens is the next cheap step.

Narrower view = finer aim

The tracker aims to about one pixel, so degrees per pixel sets the accuracy.



Recommendation. Build with the MK2 camera first. When the rig works, add the 1080p M12 board with a 6 mm lens: about 2× finer aim for ≈₹2,100–4,100. It bolts to the front plate as it is (the slots take 28–34 mm hole patterns).

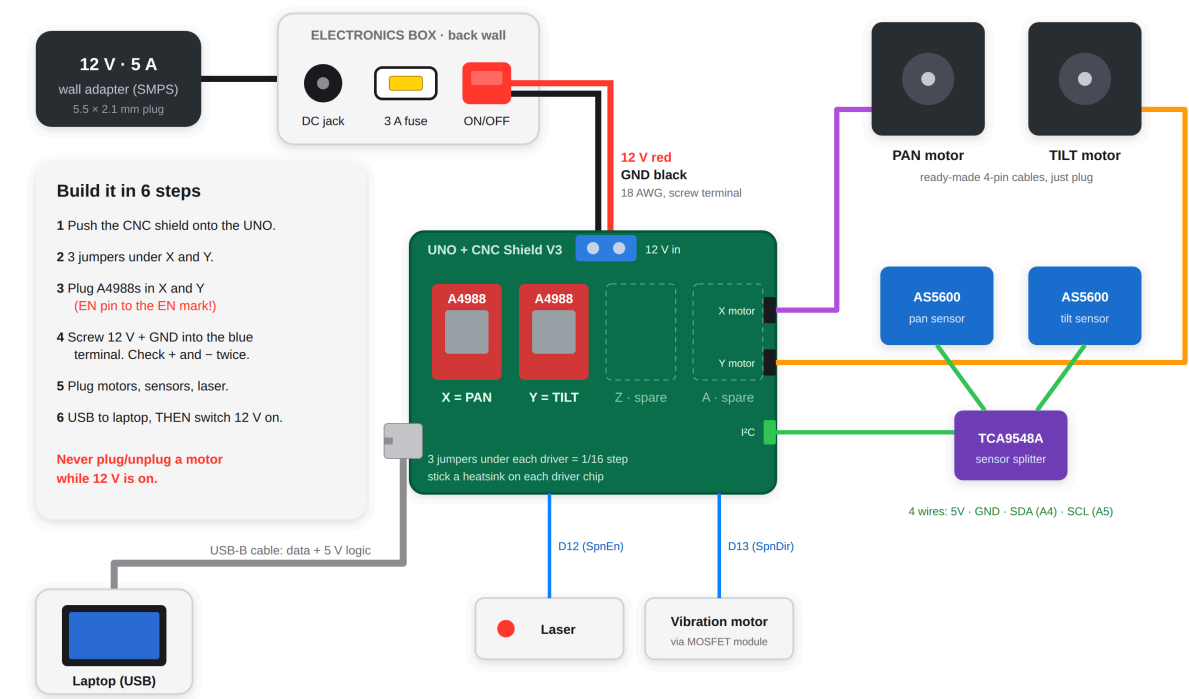
Narrow lens trade-off. A 6 mm lens sees about 51° instead of 70°, so the beacon has to be closer to the centre before the tracker can find it. The scan mode on the Live page covers that. Avoid 8 mm or longer lenses for the demo.

CAMERA	PRICE	VIEW	WHY
Keep the MK2 USB camera	Free	1280 px across about 70° 0.055° per pixel	Works today. Nothing to buy.
USB 1080p camera board with M12 lens mount (OV2710 or IMX291) + 6 mm M12 lens Best upgrade	₹1,800–3,500 + lens ₹300–600	1920 px across about 51° 0.027° per pixel	Recommended upgrade: about 2× finer aim. The 38 × 38 mm board fits the front plate. Online (Amazon: "USB camera module 1080p M12").
Logitech C270	₹1,300–1,700	1280 px across about 55° 0.043° per pixel	Easy to find locally. A little better than a basic webcam. Mount with a strap, not the front plate.
ESP32-CAM, OV7670, Raspberry Pi camera Skip	—	—	Not recommended: Wi-Fi delay, low quality, or needs a Raspberry Pi.

Wiring and power

Plan Urena · the easy wiring

No breadboard. No soldering. Every part plugs in or screws in.



No breadboard, no soldering. Every wire plugs in or screws in.

Why the old setup burned, and what stops it now

WHAT WENT WRONG	PLAN URENA
Motor current (up to 1.5 A) through breadboard strips, which are made for about 0.5 A. They get hot and arc.	12 V goes on 18 AWG silicone wire into a screw terminal on the CNC shield.
No big capacitor at the driver. Plugging in 12 V makes a voltage spike that can kill an A4988.	The CNC shield has the 100 µF capacitor built in, right beside the drivers.
A loose adapter pigtail can short.	Panel jack in the box wall, then a 3 A fuse, then a switch. A short blows a ₹5 fuse, not the drivers.
A driver running hot for a long time.	Heatsinks, a 40 mm fan in the lid, and the current set correctly (Vref below).

Pins (UNO + CNC Shield V3)

UNO	SHIELD LABEL	GOES TO
D2 / D5	X.STEP / X.DIR	Pan driver (X socket)
D3 / D6	Y.STEP / Y.DIR	Tilt driver (Y socket)
D8	EN	Both drivers on/off (LOW = on)
D12	SpnEn	Laser signal

D13	SpnDir	Vibration motor (through the MOSFET module)
A4 / A5	SDA / SCL	TCA9548A splitter → both AS5600 sensors
5V / GND	5V / GND	Sensors, splitter, laser, MOSFET module
12 V terminal	blue screw terminal	From the ON/OFF switch (red +, black –)

Clone shields sometimes label the SDA/SCL pins differently. Take a photo of yours and check before you plug the sensor wires in.

Set the motor current (Vref) once, before the motors move

1. Motors plugged in, 12 V ON, USB unplugged. Multimeter on DC volts.
2. Black probe on any GND. Red probe touching the small metal screw (potentiometer) on the driver.
3. Turn the screw with a small plastic or ceramic screwdriver until the meter shows the value below. Do both drivers.

TINY RESISTOR NEAR THE CHIP SAYS	SET VREF TO	MOTOR CURRENT
R100 (most boards)	0.64 V	0.8 A: runs cool, and the 4:1 belt gives plenty of torque
R050	0.32 V	0.8 A
R200	1.28 V	0.8 A

Formula: $V_{ref} = 8 \times R \times I$. If a motor misses steps under load, raise it by 0.1 V at a time, up to 1.0 A.

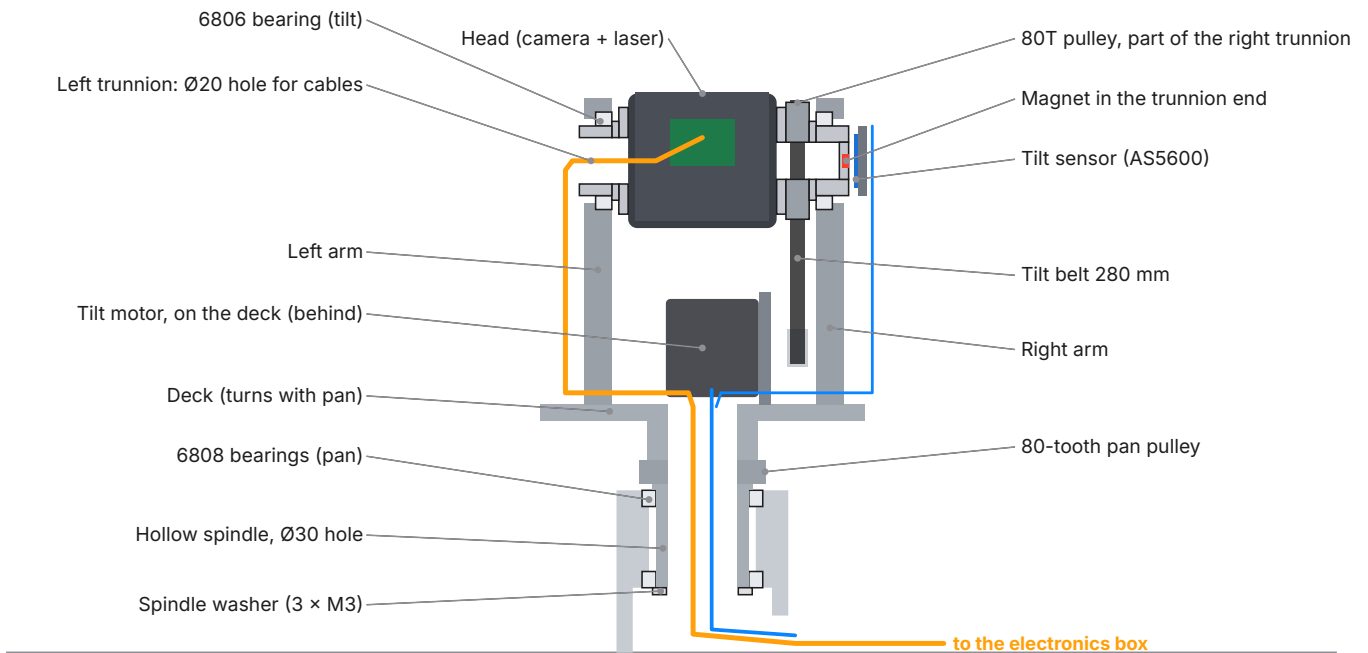
Five rules

1. Check + and – of the adapter with a multimeter before you connect it the first time.
2. Driver the right way round: its EN pin goes to the EN mark on the shield. Backwards = dead driver.
3. Never plug or unplug a motor while 12 V is on.
4. USB into the laptop first, then switch 12 V on. Switch 12 V off before unplugging USB.
5. Three jumpers under each driver = 1/16 step. The firmware expects this.

Where the cables go

Front view, cut through the middle. Orange: camera + laser. Blue: tilt motor + tilt sensor.

The tilt motor stands on the deck under the head; everything runs down the hollow pan spindle.



Assembly

The belts: 4 motor turns = 1 axis turn

Pan belt: 232 mm closed GT2, 6 mm wide

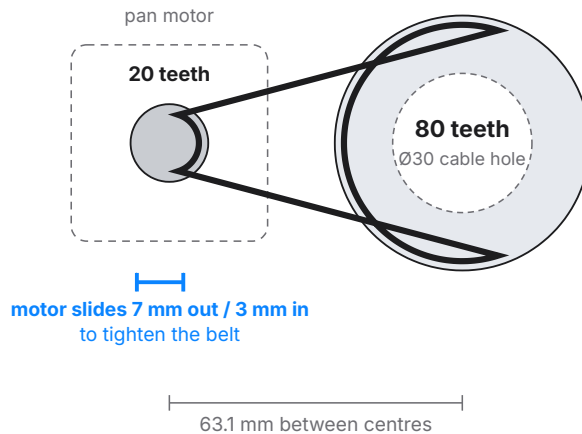
(any 227–245 mm fits, so 240 mm is fine too)

Tilt belt: 280 mm closed GT2 (276–284 mm fits)

4:1 means 4 × the torque and 4 × finer steps:

35.6 motor steps per degree of the rig.

Metal 20T pulleys on the motors, printed 80T on the axes.

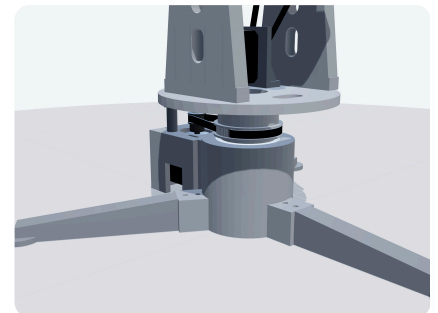


1 Check the test coupon

Bearings press into the pockets with firm thumb pressure: no hammer, and they must not fall out. The printed pins slide into the bearings with a firm push. An M3 nut drops into its hexagon; a screw threads into the small hole. If any fit is wrong, stop and tell the team: one number in the CAD fixes it.

2 Legs

Slide the three legs into the base hub. M4 × 30 bolts from the top, nuts in the hexagons underneath.

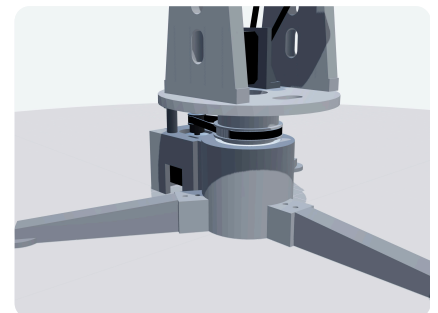


3 Pan bearings

Press one 6808 into the top of the base hub and one into the bottom (through the big hole underneath). Press on the outer ring only, with a flat block.

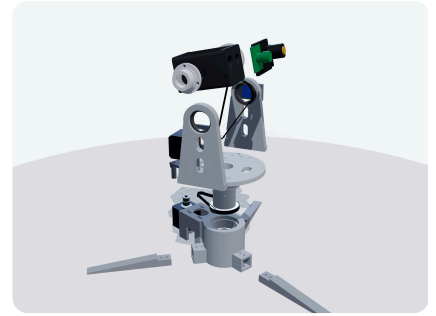
4 Pan motor

Put the 20T pulley on the motor shaft with its hub DOWN, 0.5 mm above the motor face, grub screw on the flat. Push the motor into the tower from below. Two front M3 × 8 screws, loose for now.



5 Pan turntable

Loop the 232 mm belt around the turntable's pulley. Lower the hollow spindle down through both bearings and hook the belt over the motor pulley. From underneath: the spindle washer and 3 × M3 × 12. Tighten until there is no play, but the deck still turns freely.

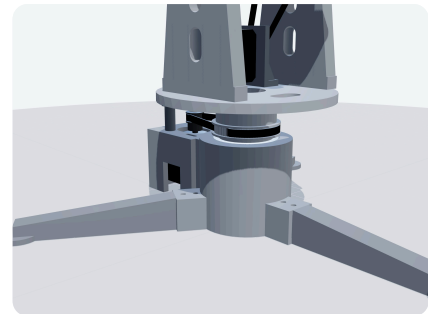


6 Pan belt tension

Slide the motor outwards until the belt makes a low "twang" when plucked, then tighten the motor screws.

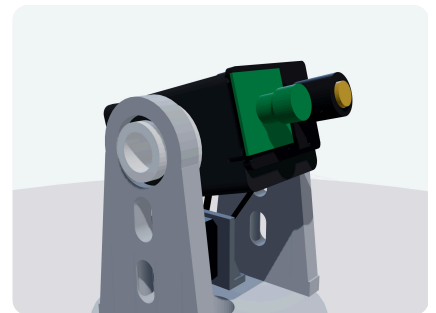
7 Pan sensor

Press the magnet into the magnet cap (flush with the top). One drop of super glue inside the cap, push it onto the motor shaft tip. Screw the AS5600 under the sensor bridge, chip facing down. Fix the bridge with the two back motor screws (M3 × 35).



8 Head

Camera board on the front plate; front plate on the head. Laser into its holder, holder onto the front plate. Left (hollow) trunnion: 3 × M3 × 10, nuts in the flange. Right trunnion (the one with the pulley): 3 × M3 × 12 from inside the head. Press the tilt magnet into the end of the right trunnion.

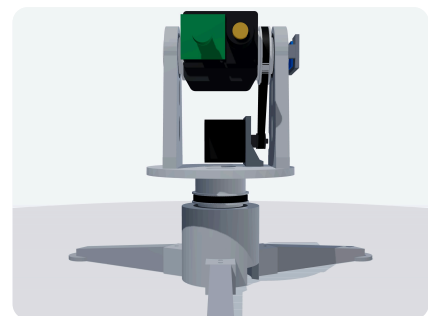


9 Tilt bearings

Press one 6806 into each arm, from the inner face (the side with the big pocket).

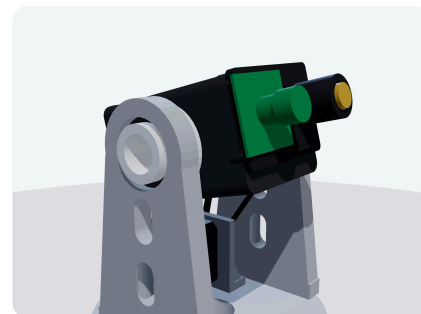
10 Head into the arms

Hold the head between the arms and slide both arms onto the trunnions. Stand the arms on the deck and bolt them (M4 × 20, nuts in the deck). Push the arms gently towards the head before tightening: no side play, but the head still tilts easily.



11 Tilt drive

Tilt motor onto its stand (4 × M3 × 10, loose) with the 20T pulley, hub towards the stand. Stand onto the deck under the head (2 × M4 × 16, nuts in the deck). Loop the 280 mm belt over the pulley on the right trunnion and the motor pulley. Slide the motor down until the belt twangs, tighten. Tilt sensor bracket on the outside of the right arm, AS5600 in it, chip facing the magnet.

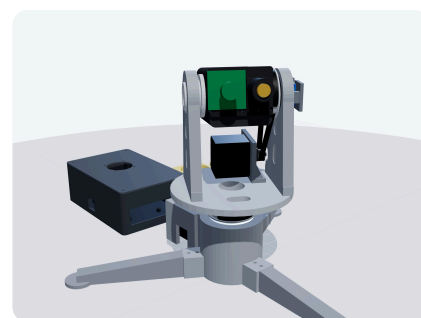


12 Cables

Camera USB and laser wires: out through the left trunnion, down the left arm (cable ties), into the hole in the middle of the deck, down the spindle, out at the bottom of the base. Tilt motor wires go straight into the deck hole (the motor sits right beside it); tilt sensor wires down the right arm. Leave a loose loop above the deck so a ±150° pan never pulls.

13 Electronics box

DC jack, fuse holder and switch into the wall. UNO on the standoffs, CNC shield on top, drivers in X and Y, jumpers in. 12 V wires: jack → fuse → switch → shield terminal. Fan in the lid. TCA9548A and the MOSFET module on their standoffs.



14 First power-on

Follow "Wiring": check polarity, set Vref, flash the firmware, run the self-test.

Screws, step by step

WHERE	SCREW	QTY
Legs → base hub	M4 × 30 + nut	6
Pan motor → base hub (front 2)	M3 × 8	2
Pan motor + sensor bridge (back 2)	M3 × 35	2
Spindle washer → turntable	M3 × 12	3
Arms → deck	M4 × 20 + nut + washer	8
Left trunnion → head	M3 × 10 + nut	3
Right trunnion → head (from inside the head)	M3 × 12	3
Tilt motor → its stand	M3 × 10	4
Tilt motor stand → deck	M4 × 16 + nut	2
Tilt sensor bracket → right arm	M3 × 12	2
Front plate → head	M3 × 12	3
Camera board → front plate	M2 or M3 × 8 + nut	4

Laser aiming screws	M3 × 8	3
AS5600 boards	M2.5 × 6	4
UNO → box	M3 × 6	4
Fan → lid	M3 × 12	4
Lid → box	M3 × 10	4

Firmware and simulation

One file: `rig_firmware_v3.ino`. The same protocol as MK2, so the Live page and the laptop software do not change.

Flash it

1. Arduino IDE 2 → Library Manager → install **AccelStepper**.
2. Open `rig_firmware_v3.ino`. Tools → Board → **Arduino Uno**. Pick the port. Upload.
Or on the Mac: `tools/rig/flash.sh urena`
3. Serial Monitor at 115200. It prints `# ZeroDrift Mk2 ready, encoders=yes, belts=4:4`. (It still says "Mk2" on purpose: that is how the Live page recognises a stepper rig.)
4. Type `!` and Enter: the self-test blinks the laser, turns pan +10° and back, and prints `# encoder saw 10.0x deg`. If it says "direction reversed", flip `PAN_ENC_DIR` in the file and upload again.
5. Live page → connect the rig → **MK2 DRIVE: "Both belts 20T → 80T (4:1)"**.

Spare board. The same file runs on the MK2 Nano: change `#define BOARD_CNC_SHIELD 1` to `0`. **Python driver:** `mk2.py` needs `steps_per_rad × 4` for the belts.

Try it first in Chrome (Wokwi, free)

1. Open wokwi.com/projects/new/arduino-uno.
2. Paste `sketch.ino` and `diagram.json` from the Wokwi folder, and add the library **AccelStepper**.
3. Press ▶. Type `P3556 T-711`: the pan motor turns 400° (= 100° of the rig, because of the belt), tilt -80°. Type `L7.0`: the red LED blinks 7 times a second. `P999999` stops at 150°: the limit works.

Tested on a PC

The firmware was compiled on a PC with fake motors, fake sensors and a fake clock, and checked **42 times on each board** (UNO + CNC shield, and Nano): travel limits, the 4:1 maths, both sensors over the full ±150° (including the sensor's 0/360 seam), skipped steps showing up, the laser at exactly 7.00 Hz, junk input, and the self-test (including a sensor mounted the wrong way round). All pass. Run it again with `bash tools/mk3/fwtest/run.sh`.

Tests

Tick these in order. Each one catches a mistake before it costs a part.

Before printing everything

- ☐ Test coupon: bearings press in firmly, pins slide into the bearings, nut fits, screw holds
-

Mechanics (belts off)

- ☐ Pan turns smoothly by hand, no wobble when you rock the deck
-
- ☐ Tilt turns smoothly, no side play in the head
-
- ☐ Belts on: a low "twang", teeth fully in the pulleys
-

Electronics (before the motors move)

- ☐ Adapter polarity checked with a multimeter
-
- ☐ Drivers the right way round (EN to EN)
-
- ☐ Three jumpers under X and Y
-
- ☐ Vref set on both drivers
-
- ☐ Fuse in, switch off, then USB, then 12 V on
-

Firmware

- ☐ Banner says encoders=yes, belts=4:4
-
- ☐ Self-test: "encoder saw ~10.00 deg"
-
- ☐ "P3556 T-711" then "?" gives about S 3556 -711 E 0
-
- ☐ After 10 minutes: drivers warm but touchable, E and C still agree
-

Tracking

- ☐ Laser aimed at the camera centre (Live page → SET AIM)
-
- ☐ Live page drive menu = "Both belts 20T → 80T (4:1)"
-
- ☐ Locks on the beacon, ignores the decoy
-
- ☐ Pan $\pm 150^\circ$ with no cable pulling
-